

NOVEMBER 2, 2025

DEPARTURE DELAYS

Analysis

Report

APPLIED STATISTICAL INFERENCE & EXPERIMENTAL DESIGN, DATA 5300

INTRODUCTION

Technique

We aim to analyze the relationship between departure delay times for United Airlines NYC flights and each of the following variables:

- Time of day
- Time of year
- Temperature
- Wind speed
- Precipitation
- Visibility

Based on our findings, we will prepare a report that utilizes exploratory data analysis methods, regression analysis, and permutation tests to analyze variables that affected departure delays for the year of 2013.

Although our problem statement focuses on one variable in the end; we want to incorporate other variables into our analysis so that we can better understand **which one** impacts flight delays the most.

Problem Statement

The purpose of the assignment was to perform work using the data science methodology to answer the question: **What variable causes the greatest impacts on departure delays for United Airlines flights?**

Analytic Approach

These conclusions will be decided based on what we find when working through our descriptive and regression analysis of the data, then tested for randomization in our permutation testing.

The exploratory data analysis segment of this process will help us visualize the relationships between departure delays and the 6 selected flight variables, while the regression analysis segment will help us test these findings statistically.

In the end, we will test for randomization and use a **Diagnostic Approach** to make inferences about which variable had the greatest impact on 2013 United Airlines NYC flight delays.

METHODOLOGY

Data Sources

This project utilizes four main data sets from the **nycflights13** R package.

The first data set, **weather**, contains values from the variables we are going to analyze further. Like the departure airport, time of day, time of year, visibility, precipitation level, and wind speed.

The second data set, **airlines**, provides details about each airline, including the carrier airline code and airline name.

The third data set, **airports**, contains details about each airport, like the faa code, name, altitude, latitude/longitude, altitude(ft), timezone, and daylight-saving indicators .

The fourth data set, **flights**, contains details about the United Airlines flights in and out of NYC for the year of 2013, such as the airline and carrier names, airport information, plane identification codes, and weather origin.

Together, these datasets allow for the analysis of relationships between departure delays and the 6 variables we aim to analyze.

Initial Analysis

We relied on the following installation packages, outside of **nycflights13**, to import and analyze the variables in each data set:

- **Ggplot2**: Imported for data visualization
- **Dplyr**: Used for data manipulation in R. Such as the filter, select, mutate, arrange, and summarize functions
- **Tidyr**: Used for tidying and reshaping data
- **Tibble**: Installed modern data frames for R
- **Forcats**: Imported for categorical/factor data manipulation
- **Lubridate**: Used to work with date and time
- **Infer**: Used for creating statistical inferences based on simulations

Once all packages and libraries were installed, we first started off with an initial analysis of all four data sets. Instead of working through each attribute for the dataset, we created a function that outputs all the information we want for each dataset; which we called **describe_dataset()**. With this information, we were able to identify data types, size of each data set, and count missing values.

This helped us gain a better understanding of precisely what data we had to work with, and what changes we wanted to implement before building out visuals and running any tests. For example, this process helped us identify which key variable we wanted to merge the four data sets on, and which variables to exclude in the final merge.

Data Cleaning

We decided to move forward and rename variables to improve readability and uniformity across all four data sets, condense down each dataset to the variables we need to analyze the relationships, and create 5 additional variables:

- We added **'time_of_day'** to the **flights** data set to display Morning, Afternoon, Evening, or Night.
- We added **'time_of_year'** to the **weather** data set to display Fall, Winter, Spring, or Summer
- We added **'time_departed'** to the **flights** data set to display the time each flight departed their origin in date-time format.
- We added **'time_arrival'** to the **flights** data set to display the time each flight arrived at their destination in date-time format.

We did not want to alter the original data sets in case we wanted to reference them later, so the data sets we chose to incorporate in our further analysis include the original **'airlines'** and **'airports'** data sets, and two newly created data set that house all the adjustments, changes, and new variables – which we called **'weather_new'** and **'flights_new'**.

Data set summary information, for both original and new data sets, can be viewed in our [Google Colab Notebook](#) under the *Initial Analysis* and *Data Cleaning* segments.

Data Prepping

Data Filtering

To analyze all the variables against one another, we needed all the data in one data set. Before merging all four data sets together we filtered the main **flight** dataset to include only flights operated by United Airlines (UA) and verified that all data types and formatting was uniform across all the data sets; this new filtered data set is called '**ua_flights**'.

When we tried to merge the United Airlines flight data with the **weather** data, R kept crashing and said we ran out of memory. Originally, our flight data had around 58,665 flights, and our weather data had around 26,115 rows of weather readings, each airport (EWR, JFK, LGA) had thousands of hourly records throughout the year.

When we merged by **airport**, R tried to attach every single weather record to every flight from that airport creating millions of rows and filling up memory. We decided to keep only **one weather record for each airport and season** (Winter, Spring, Summer, Fall), which reduced 26,115 rows to just 12 (3 airports × 4 seasons).

The data set now contains rows of flight data (delay time, time of day, etc.), in addition to the airport name, and seasonal weather information (temperature, wind speed, precipitation, visibility). This makes the dataset easier to analyze.

Joining the Data Sets

Now, onto the merge! We first created a new dataset - called '**ua_base**' - by merging the filtered United Airlines flights with the airlines and airports data sets. Below commands show that each airport had thousands of hourly weather records:

```
table(weather_new$origin_airport_faa)

  EWR   JFK   LGA 
8703 8706 8706 

table(ua_base$origin_airport_faa)

  EWR   JFK   LGA 
46087 4534 8044
```

When we merged using only the airport column, R combined every flight with every weather row from the same airport, causing massive duplication and memory overflow.

Next, we reduced the weather dataset by keeping only one record per airport and season (Winter, Spring, Summer, Fall). This way, each airport only had four weather rows instead of thousands:

```
weather_unique <- weather_new %>%
  distinct(origin_airport_faa, time_of_year, .keep_all = TRUE)

head(weather_unique)
```

A tibble: 6 x 6

origin_airport_faa	temperature	wind_speed	precipitation	visibility	time_of_year
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
EWB	39.02	10.35702	0	10	Winter
EWB	37.04	4.60312	0	10	Spring
EWB	78.08	6.90468	0	10	Summer
EWB	80.06	5.75390	0	10	Fall
JFK	39.02	12.65858	0	10	Winter
JFK	39.02	10.35702	0	10	Spring

After simplifying the weather data, we successfully merged it with ‘**ua_base**’ to create a final dataset called ‘**ua_combined**’, which contain the following variables:

```
colnames(ua_combined)
```

'year' · 'departure_delay_minutes' · 'arrival_delay_minutes' · 'airline_carrier_code' · 'origin_airport_faa' · 'destination_airport_faa' · 'air_time' · 'time_of_day' · 'time_departed' · 'time_arrival' · 'airline_name' · 'airport_name' · 'lat' · 'lon' · 'alt' · 'tz' · 'dst' · 'tzone' · 'temperature' · 'wind_speed' · 'precipitation' · 'visibility' · 'time_of_year'

Imputing Missing Values

An important part of analyzing the data is to make sure that all missing values are accurately identified and imputed. We confirmed a total of **5,740 missing entries** and identified the location where these occurred in our final data set.

We imputed all missing *numerical* values by replacing them with the calculated mean across the column they reside in, and imputed all missing *categorical* values with ‘NAs’.

We verified the results of our imputation where all missing values were replaced/imputed, except in variables **time_departed** and **time_arrival**. It is important to note that these continue to display ‘NAs’ for missing values, as this is the most accurate representation for unrecorded or cancelled flights.

ANALYSIS

Regression Analysis

We decided to run a **multiple regression model** to analyze how United Airlines departure delays are affected by various weather variables (temperature, wind speed, precipitation, visibility, time of day, time of year). Doing this will help us statistically test which of these variables are **strong predictors** in influencing departure delays.

Doing this **before the EDA** helps us narrow down to a subset of variables when creating visuals, like plots and graphs, based off what does and does not show impact on delay times.

```

Call:
lm(formula = departure_delay_minutes ~ temperature + wind_speed +
    precipitation + visibility + time_of_day + time_of_year,
    data = ua_combined)

Residuals:
    Min       1Q   Median       3Q      Max
-39.96 -15.09  -8.71   -0.47  470.21

Coefficients: (2 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -16.38606    3.00660   -5.450 5.04e-08 ***
temperature     0.38169    0.03883    9.829 < 2e-16 ***
wind_speed    -0.15011    0.05012   -2.995 0.00274 **
precipitation      NA         NA      NA      NA
visibility      NA         NA      NA      NA
time_of_dayEvening  8.64888    0.19085   45.317 < 2e-16 ***
time_of_dayMorning -8.32353    0.17275  -48.181 < 2e-16 ***
time_of_dayNight   7.80080    0.51586   15.122 < 2e-16 ***
time_of_yearSpring 16.00571    1.64031    9.758 < 2e-16 ***
time_of_yearSummer  0.90259    0.22690    3.978 6.96e-05 ***
time_of_yearWinter 16.14118    1.67800    9.619 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 34.8 on 234651 degrees of freedom
Multiple R-squared:  0.03919, Adjusted R-squared:  0.03915
F-statistic: 1196 on 8 and 234651 DF, p-value: < 2.2e-16

```

In the output displayed above, variables such as **temperature** and **wind speed** illustrate meaningful effects, showing that changes in weather conditions can influence departure times. On the other hand, **precipitation** and **visibility** were automatically excluded from the model because of multicollinearity, which means their effects were already represented by other predictors like temperature and time of year.

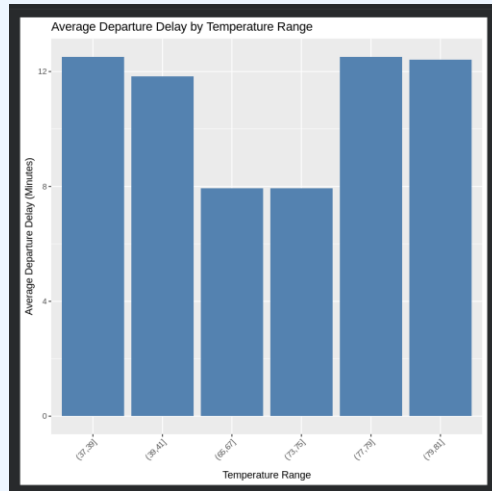
Among the time-related factors, **both time of day and time of year were strong indicators** of delay variation. This tells us that flights scheduled during certain **times of the day** and in **particular seasons** generally experienced longer delays than others.

The Adjusted R-squared value of **0.039** indicates that the model explains **approximately 3.9% of the variation in departure delays**. Although this accounts for only a small portion of the total variability, such a result is typical for real-world flight data, where many random and operational factors influence delays. Nevertheless, the results demonstrate/ that several predictors have a statistically significant relationship with flight delays – but more specifically, time of day and time of year!

Exploratory Analysis

In this segment of our analysis process, we aimed to gain a descriptive understanding of the relationships between **departure delay time** and variables that were proven to be **strong predictors** in the above regression analysis.

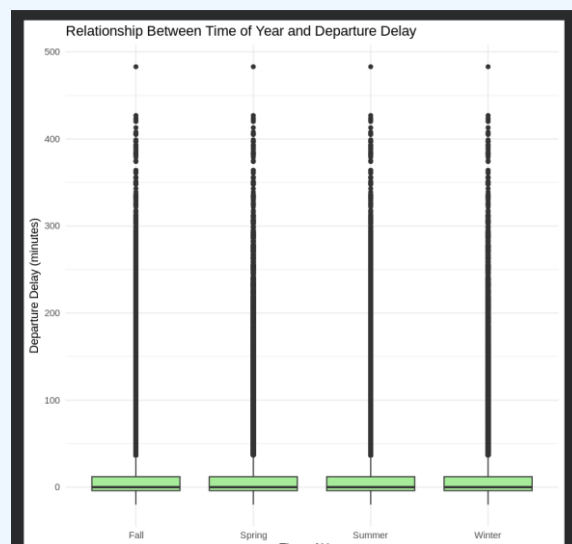
Relationship Between Temperature and Delay



Summary:

Analysis of the relationship between temperature and average departure delays shows that there is no smooth or continuous trend linking the two. The highest average delays, around 12 minutes, occur during periods of extreme cold and extreme heat, while the lowest delays are observed in milder, warmer conditions. Overall, the difference in average delay between the best and worst temperature ranges is only about four minutes, indicating that temperature is not a major driver of delays compared to other factors. In conclusion, while very cold or very hot weather may slightly increase the likelihood of delays, the effect is minor.

Relationship between Season (time_of_year) and Delay

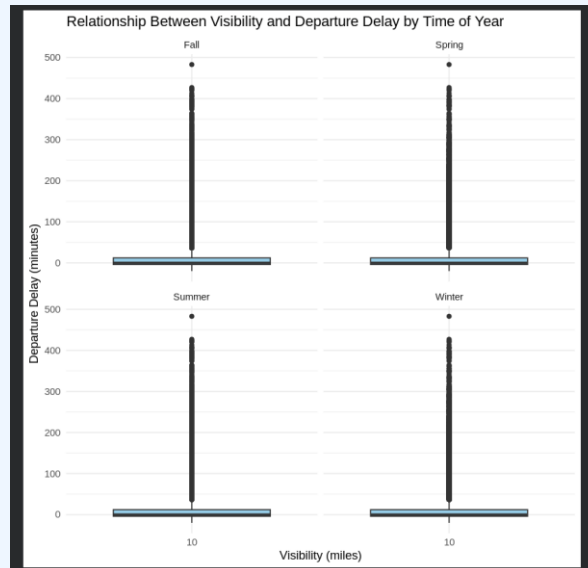


Summary:

Analysis of departure delays across the four seasons shows that the most common delay time, represented by the median, is near zero, indicating that flights are typically on time regardless of season. However, the variation in delays is largely driven by extreme, long delays that occur in every season. Summer and Winter show a slightly higher frequency of these longest delays compared to Fall.

and Spring. While season does not appear to affect routine on-time performance, it may influence the likelihood of the most severe delay events. In conclusion, season is not a major factor in everyday delays but is associated with the highest-impact delay occurrences.

Relationship between Visibility and Delay for Each Season



Summary:

Analysis of departure delays in relation to visibility indicates that changing visibility from 10 miles (clear) to 0 miles (very poor) has little effect on the most common (median) delay across all seasons. Even under clear conditions, the data show a substantial number of extreme delays, suggesting that outliers persist regardless of visibility. While very poor visibility can contribute to some long delays, the overall distribution of delays, including the frequency of extreme events, remains largely similar to conditions of high visibility. This indicates that visibility alone is not the primary driver of widespread long delays at United Airlines, and that other factors, such as airport congestion or operational challenges, are likely more influential.

PERMUTATION TESTING

Although the RA and EDA provide sufficient information to help us answer our original problem statement, there is one more test we want to apply to test for accuracy in our analysis. The regression analysis we did previously tested for “strong indicators” of departure delay times, but a **permutation test** - also called a randomization test - will test whether an observed difference between variables is **statistically significant**. It works by using the data to build a sampling distribution under the null hypothesis.

In other terms, we have already identified through the RA which of the 6 variables has the greatest effect on departure delays, but we want to see if this is truly accurate or just by random chance.

To do this we performed a permutation test for two hypothetical questions:

- **Hypothesis 1:** Does `time\_of\_day` affect departure delays?
- **Hypothesis 2:** Does `time\_of\_year` affect departure delays?

Hypothesis 1, Code

```
library(dplyr)
library(infer)

# Filter data and ensure no missing values
ua_timeofday <- ua_combined %>%
  filter(!is.na(time_of_day), !is.na(departure_delay_minutes))

# Observed F-statistic, variation in delays (since time_of_day has >2 groups)
obs_stat_day <- ua_timeofday %>%
  specify(departure_delay_minutes ~ time_of_day) %>%
  calculate(stat = "F")

# Permutation test (randomizing time_of_day labels)
# shuffle 500 times and recalculate
set.seed(123)
perm_day <- ua_timeofday %>%
  specify(departure_delay_minutes ~ time_of_day) %>%
  hypothesize(null = "independence") %>%
  generate(reps = 500, type = "permute") %>%
  calculate(stat = "F")

# Compute p-value
p_value_day <- perm_day %>%
  get_p_value(obs_stat = obs_stat_day, direction = "greater")

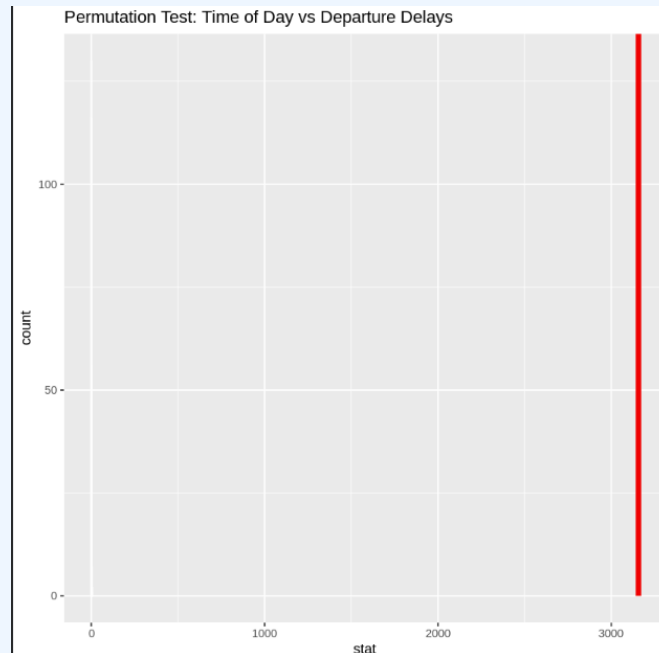
p_value_day
```

If this first hypothetical we tested for how much variation in delays can be explained by differences in **time_of_day**.

We then shuffled the **time_of_day** labels 500 times and recalculated that same statistic each time, which built a null distribution of “random” differences — what we would expect if time of day didn’t matter at all.

Hypothesis 1, Output

Our p-value ended up outputting to zero! This is a good sign. This p-value is less than 0.002 (1/500 = 0.002), thus indicating a **large statistical significance** for this variable, meaning that the observed differences in delays across morning, afternoon, evening, and night flights are **very unlikely** to be due to random chance.



This was also proven visually in how the display above shows **time_of_day** versus departure delays. The red line (F-statistic) is way off to the right, which means that none of the shuffled data produced a large effect of randomness.

Hypothesis 2, Code

```
# Filter for complete data
ua_timeofday <- ua_combined %>%
  filter(!is.na(time_of_year), !is.na(departure_delay_minutes))

# Observed F-statistic
obs_stat_year <- ua_timeofday %>%
  specify(departure_delay_minutes ~ time_of_year) %>%
  calculate(stat = "F")

# Permutation test (randomizing time_of_year labels)
set.seed(123)
perm_year <- ua_timeofday %>%
  specify(departure_delay_minutes ~ time_of_year) %>%
  hypothesize(null = "independence") %>%
  generate(reps = 500, type = "permute") %>%
  calculate(stat = "F")

# Compute p-value
p_value_year <- perm_year %>%
  get_p_value(obs_stat = obs_stat_year, direction = "greater")

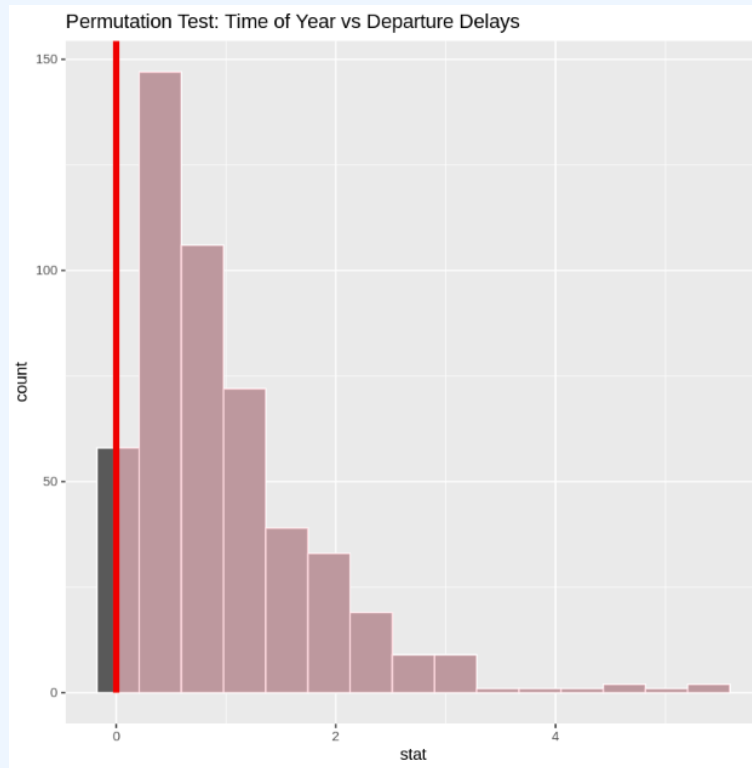
p_value_year
```

Following the same observations as above, this tested the actual F-statistic for our original data — how much variation in delays can be explained by differences in **time_of_year**.

We then shuffled the **time_of_year** labels 500 times and recalculated that same statistic each time. This built a null distribution of “random” differences — what we would expect if time of year didn't matter at all.

Hypothesis 2, Output

This p-value outputted to equal 1, thus indicating that there is **no statistically significant relationship between the season and the average departure delay** times. This suggests that the observed differences between Winter, Spring, Summer, and Fall were consistent with what could occur by **random variation**.



Also confirmed visually, this graph above shows the **time_of_year** versus departure delays, where the red line(F-statistic) sits inside the main body of the histogram. This suggests that there were many randomized results that were extreme/just as extreme as the real one.

Permutation Test Summary

Out of the variables that were determined to be “strong predictors” of departure delays, it is observed that **time_of_day** is the only variable to display a differences in departure delay times that is statistically significant, and not due to random chance.

DISCUSSION

For 2013 United Airlines NYC flights, we found that the strongest predictors of departure delays were **time_of_day** and **time_of_year**, but when tested for randomness – we found that **time_of_day** was the only variables that displayed any level of statistical significance.

So, to answer our original question - what variable has the greatest effect on departure delays? - our suggested solution would be **time of day** (morning, afternoon, evening, night).

To further expand on this conclusion, it is important to emphasis that this is just based on the analysis of a very specific subset of data – United Airlines NYC for the year of 2013. If we wanted to gain a better understanding of variable effects on United Airlines' departure delays, we would want to analyze data across multiple origins and years.

For example, we could analyze the relationship of departure delays for NYC Unites Airlines flights across the time span of 2000 to 2020, instead of just in 2013. This would allow us to see a bigger picture in variables like **time_of_year** (winter, spring summer, fall).

Just like our permutation test suggested, the differences we observed in the departure delays for each season is mostly likely due to random chance. Logically, we can consider the fact that this is not fully true. Time of year must affect departure delays, right?

Expanding this analysis further by analyzing this across multiple years allows us to pick up any potential trends.